

1. Find the Second Largest Element (without sorting)

Input:

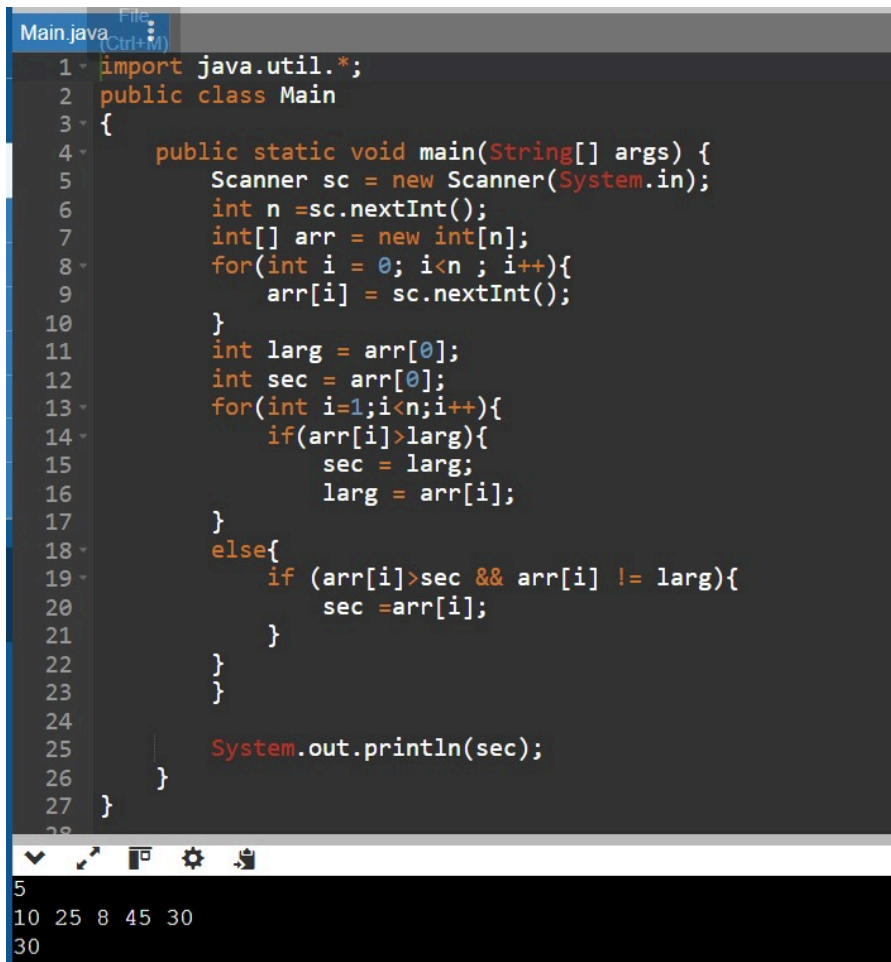
5

10 25 8 45 30

Output:

Second Largest = 30

code:

A screenshot of a Java IDE window titled 'Main.java'. The code defines a 'Main' class with a 'main' method. It uses a 'Scanner' to read an integer 'n' and an array of 'n' integers. It then iterates through the array to find the largest and second largest elements. The output is printed as 'Second Largest = 30'. Below the code editor, the input '5' and the array '10 25 8 45 30' are shown, followed by the output '30'.

```
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         for(int i = 0; i < n; i++){
9             arr[i] = sc.nextInt();
10        }
11        int larg = arr[0];
12        int sec = arr[0];
13        for(int i = 1; i < n; i++){
14            if(arr[i] > larg){
15                sec = larg;
16                larg = arr[i];
17            }
18            else{
19                if (arr[i] > sec && arr[i] != larg){
20                    sec = arr[i];
21                }
22            }
23        }
24
25        System.out.println(sec);
26    }
27 }
```

5
10 25 8 45 30
30

2. Find the Second Smallest Element

Input:

6
9 2 7 5 1 8

Output:

Second Smallest = 2

Code:

```
Main.java
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         for(int i = 0; i < n; i++){
9             arr[i] = sc.nextInt();
10        }
11        int larg = arr[0];
12        int sec = arr[0];
13        for(int i = 1; i < n; i++){
14            if(arr[i] < larg){
15                sec = larg;
16                larg = arr[i];
17            }
18            else{
19                if (arr[i] < sec && arr[i] != larg){
20                    sec = arr[i];
21                }
22            }
23        }
24
25        System.out.println(sec);
26    }
27 }
28
```

6
9 2 7 5 1 8
2

3. Move All Zeros to the End

Input:

1 0 5 0 8 2 0 6

Output:

1 5 8 2 6 0 0 0

Code:

```
Main.java :
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args) {
5         Scanner sc =new Scanner(System.in);
6         int[] arr = {1,0,5,0,8,2,0,6};
7         int count=0;
8         for(int i=0;i<arr.length-1;i++){
9             if(arr[i] != 0){
10                 System.out.print(arr[i]+" ");
11             } else{
12                 count++;
13             }
14         }
15         for(int i =0;i < count;i++){
16             System.out.print("0"+" ");
17         }
18     }
19 }
20
21
22
23
```

1 5 8 2 0 0 0

...Program finished with exit code 0
Press ENTER to exit console.

4. Rotate Array to the Left by One Position

Input:

10 20 30 40 50

Output:

20 30 40 50 10

Code:

```
Main.java (Ctrl+M)
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         for (int i = 0; i < n; i++) {
9             arr[i] = sc.nextInt();
10        }
11        int num = arr[0];
12        for(int i = 1;i<n;i++){
13
14            System.out.print(arr[i]+" ");
15        }
16        System.out.println(" "+num);
17    }
18 }
```

5
10 20 30 40 50
20 30 40 50 10

5. Rotate Array to the Right by One Position

Input:

10 20 30 40 50

Output:

50 10 20 30 40

Code:

```
Main.java
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args) {
5         int[] arr = {10, 20, 30, 40, 50};
6         int first = arr[arr.length-1];
7         System.out.print(first+" ");
8         for(int i=0; i<arr.length-1; i++){
9             System.out.print(arr[i]+" ");
10        }
11    }
12 }
13
```

50 10 20 30 40

...Program finished with exit code 0
Press ENTER to exit console.

6. Find Missing Number (1 to N)

Input:

1 2 4 5 6

Output:

Missing Number = 3

Code:

```
Main.java
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args) {
5         int[] arr={1,2,4,5,6};
6         int n = 1;
7         for(int i =0;i<arr.length;i++){
8             if(arr[i] != n){
9                 System.out.println("Missing Number =" + n);
10                return;
11            }
12            n++;
13        }
14    }
15 }
16
17
18
```

input

Missing Number =3

7. Find the Element that Appears Only Once

(All other elements appear twice.)

Input:

2 5 7 5 2 8 8

Output:

Unique Element = 7

Code:

```
main.java
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner (System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         //int count=0;
9         for(int i = 0; i < n; i++){
10             arr[i] = sc.nextInt();
11         }
12         for(int i=0;i<n;i++){
13             int count=0;
14             for(int j=0;j<n;j++){
15                 if(arr[i]==arr[j]){
16                     count++;
17                 }
18             }
19             if(count==1){
20                 System.out.println("unique number:"+ arr[i]);
21                 break;
22             }
23         }
24     }
25 }
```

input

```
7
2 5 7 5 2 8 8
unique number:7
```

8. Find the First Repeating Element

Input:

4 2 7 8 2 5 4

Output:

First Repeating Element = 4

Code:

```
Main.java
5
6     Scanner sc = new Scanner(System.in);
7
8     int n = sc.nextInt();
9     int arr[] = new int[n];
10
11     for(int i = 0; i < n; i++) {
12         arr[i] = sc.nextInt();
13     }
14
15     for(int i = 0; i < n; i++) {
16
17         int count = 0;
18
19         for(int j = 0; j < n; j++) {
20
21             if(arr[i] == arr[j]) {
22                 count++;
23             }
24         }
25
26         if(count > 1) {
27             System.out.println("First Repeating Element = " + arr[i]);
28             break;
29         }
30     }
31 }
32 }
```

input

```
7
4 2 7 8 2 5 4
First Repeating Element = 4
```


9. Find the First Non-Repeating Element

Input:

2 3 4 2 3 6 7

Output:

First Non-Repeating Element = 4

Code:

```
Main.java
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         int n = sc.nextInt();
9         int arr[] = new int[n];
10
11         for(int i = 0; i < n; i++) {
12             arr[i] = sc.nextInt();
13         }
14
15         for(int i = 0; i < n; i++) {
16
17             int count = 0;
18
19             for(int j = 0; j < n; j++) {
20
21                 if(arr[i] == arr[j]) {
22                     count++;
23                 }
24             }
25
26             if(count == 1) {
27                 System.out.println("First Non-repeating Element = " + arr[i]);
28                 break;
29             }
30         }
31     }
32 }
```

input

7
2 3 4 2 3 6 7
First Non-repeating Element = 4

...Program finished with exit code 0
Press ENTER to exit console.

10. Check if Array is Sorted

Input:

5 10 15 18 20

Output:

Array is Sorted

Code:

```
Main.java :
1 import java.util.*;
2
3 public class Main {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         int n = sc.nextInt();
10        int arr[] = new int[n];
11
12        for(int i = 0; i < n; i++) {
13            arr[i] = sc.nextInt();
14        }
15
16        boolean sorted = true;
17
18        for(int i = 0; i < n - 1; i++) {
19
20            if(arr[i] > arr[i + 1]) {
21                sorted = false;
22                break;
23            }
24        }
25
26        if(sorted) {
27            System.out.println("Array is Sorted");
28        } else {
29            System.out.println("Array is not Sorted");
30        }
31    }
32}
```

5
5 10 15 18 20
Array is Sorted

...Program finished with exit code 0
Press ENTER to exit console.